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Stream Data Analytics for Network Attacks' Prediction

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Abstract

Nowadays stream data, flowing over the modern networks between disparate data sources, become the norm. The broadband Internet, the Internet of Things (IoT) and cloud computing require to analyze the data from streams to make data-driven decisions in real time. In today's world of more complex and increasing in the number of network attacks, one of their most important data is the data from network security (NS) tools, ensuring their secure and resilient operations and uninterrupted provision of services to its users. At Gartner Data & Analytics Summit-2019 augmented analytics and data management, as well as continuous intelligence and explainable artificial intelligence were indicated among the top trends in data and analytics technology that have significant disruptive potential over the next 3-5 years. In practice, complexities of the modern attack scenarios often make it difficult for NS administrators to understand the current NS-related status and to recognize emerging patterns of attacks in a vast amount of raw data before they make a substantial impact. To benefit from the NS-related stream data, businesses require powerful analytics tools for ingesting and processing it. There are four consequent levels of analytics maturity - namely descriptive, diagnostic, predictive, prescriptive. In this paper, a simplified NS-related stream data architecture, suitable for predicting attacks against network assets and services provided is proposed. In turn, MITRE ATT&CK Matrix is proposed as a source for attacks' Indicators of Compromise (IoCs) development.

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1. Introduction

Nowadays stream data, flowing over the modern networks between disparate data sources, become the norm. Data streaming can be imagined as the continuous transfer of data in motion at a steady, high-speed rate [1]. The broadband Internet, the Internet of Things (IoT) and cloud computing require to analyze the data from streams to convert it into actionable insights and to make data-driven decisions in real time.

In today's world of more complex and increasing in the number of network attacks, following the data of organizations' main business processes, one of their most important data is the data from network security (NS) tools, ensuring their secure and resilient operations and uninterrupted provision of services to its users.

At Gartner Data & Analytics Summit (February 18-19, 2019, Sydney, Australia) augmented analytics and data management, as well as continuous intelligence and explainable artificial intelligence were indicated among the top trends in data and analytics technology that have significant disruptive potential over the next 3-5 years [2]. The very challenge of too much digital data together with increasingly powerful processing capabilities has created an unprecedented opportunity. According to D.Feinberg, Vice President at Gartner, "the size, complexity, distributed nature of data, speed of action and the continuous intelligence required by digital business means that rigid and centralized architectures and tools break down... The continued survival of any business will depend upon an agile, data-centric architecture that responds to the constant rate of change." "As digital businesses become inundated with data, augmented analytics is crucial for presenting to operational users only what is important for them in the context to act upon at that moment," J.Hare, a Gartner analyst, added to that.

In practice, complexities of the modern attack scenarios often make it difficult for NS administrators to understand the current NS-related status and to recognize emerging patterns of attacks in a vast amount of raw data before they make a substantial impact. To benefit from the NS-related stream data, businesses require powerful analytics tools for ingesting and processing it like popular Amazon Kinesis Firehose (an Amazon Web Service, which is capable of processing hundreds of terabytes per hour from high volumes of streaming data from sources such as operating logs, financial transactions and social media feeds, https://aws.amazon.com/kinesis/?nc1=h_ls), Apache Flink (a distributed data processing platform for use in big data applications, primarily involving analysis of data stored in Hadoop clusters, <https://flink.apache.org/>), Apache Kafka (a full-scale distributed streaming platform, capable of handling trillions of events a day) and others. In the above case, after processing by these tools information acquired will allow to get the full picture and to notify NS administrators about the emergence of one or more of a huge variety of possible problems substantially earlier than was possible before, and to enable pragmatic prioritization of response efforts according to the significance of the detections.

There are currently multiple consequent levels of analytics maturity - namely descriptive, diagnostic, predictive, prescriptive. Thus, the purpose of the paper is to work out a simplified NS-related stream data architecture, suitable for predicting attacks against network assets and services provided. In turn, MITRE ATT&CK Matrix is proposed as a source for attacks' Indicators of Compromise (IoCs) development.

2. Related Work

Stream data processing can be considered big data information technology (IT) [3]. The criteria for determining the difference between the big data IT and traditional IT are three main "Vs": volume – very large volumes of data, velocity – very high data transfer rate, variety – weak structured data, which is primarily understand as data structure irregularity, difficulty of extracting homogeneous data from a stream and identifying some correlations. Four additional "Vs" – veracity, variability, value, visibility – can be added to them [4]. At present, stream data processing combines many terms historically having some differences: streaming analytics, real-time analytics, real-time streaming analytics, event processing, etc. [5].

Applying real-time big data technologies to security analytics means stream processing of network flows, data packets and metadata looking for anomalous/suspicious network activities that provide a strong indication of a security incident in progress requiring response actions instantaneously or with as little delay as possible. Multiple vendors, including Cisco (they announced an adaptive and proactive approach to security even 15-years ago), ISC8, 21CT, Click Security, Hexis Cyber Solutions, IBM, Lancope, Leidos, LogRhythm, Netskope, RSA Security, Solera Networks and others), are known for their advantages in this area [6]. But a lot of obstacles such as too many

independent tools that aren't integrated together, too many false positive alerts, too many manual processes in attack's detection and a lack of adequate staffing still slow down the process of widespread use of these technologies for proactive protection of modern networks [6].

3. NS-related data streams

Regarding this research, big data technology will be used to query continuous NS-related data streams in order to detect conditions and anomalies in systems', users', applications' and devices' behaviour as quickly as possible, within a small time period from the time (it varies from few milliseconds to minutes) of receiving the data. The NS-related data should be considered in a particular context. It is a must to maintain the recorded relationships of every file execution, file modification, registry modification, network connection, executed binary in your environment, etc. For example, correlating logs with recent event history can be used to identify and flag possible unexpected side effects of security devices' configuration changes or upgrades.

The additional streams of NS-related data come not just from the DNS servers, proxy servers, security endpoint devices, and telemetry data sources, but from Threat Intelligence Feeds as well. It describes characteristics of network traffic, current configurations of network devices, application and network services functioning, an activity of individual users, as well as contain extensive amounts of self-diagnostic data from network devices and built-in test systems, machine-generated logs, e-mails, web-based content, user's queries' results, data streams from video surveillance systems and access control and alarm systems, website's clickstreams and another kinds of time-series data, metadata, business processes' inputs and outputs, enterprise's internal documents, and analytical data for many years of its existence, etc.

The main sources of NS-related raw stream data are technical, software and hardware security endpoint devices, as well as telemetry data sources used for NS monitoring and operation control. All these data should not be considered as a combination of separate data elements. Moreover, it is a data stream with the following unique features: huge or possibly infinite volume, dynamically changing, flowing in and out in a fixed order, demanding real-time response, etc. Real-time ingestion and stream data processing will create new opportunities for big data analytics and will enable rapid threat prediction. For example, with stream data processing, you can receive an alert when the volume of traffic has overcome the threshold value, querying data streams coming from one network segment to another one.

To process all these data to predict possible network attacks an appropriate data architecture should be proposed.

4. A simplified NS-related stream data architecture

Fig. 1 illustrates our approach for real-time analytics, leveraging fast operational database by combining operational and analytical workloads in one place to improve decision-making while processing data for analysis extracted from the data lake. Further, all the terms used are explained.

Raw data refers to the relatively unprocessed stream of big data. We propose to store it in a data lake consolidating data silos. The "data lake" concept was introduced by J.Dixon in 2010 [7]. It can be imagined as a large data pool to bring in all of the historical data accumulated and new data in near real-time into one single place. The schema and data requirements to data lakes are not defined until the data is queried (in contrast to data warehouses) [4].

Exploratory analytics is designed to perform initial investigations on raw data to discover patterns and anomalies, as well as to check hypothesis and assumptions with the help of summary statistics.

Processed data is a subset of the raw data that has been processed, perhaps to create aggregations and categories (e.g., users' profiles, software run daily) and perform some augmentations. They are smaller in volume than the raw data, but still requires specialized tools and expert skills to extract data for analysis.

Data for analysis is presented in a format ready for further analysis and has managed access.

Data presentation includes various reports, dashboards, and templates that surface the processed by analytical systems data to users in preconfigured formats and visual layouts for easy understanding and decision-making.

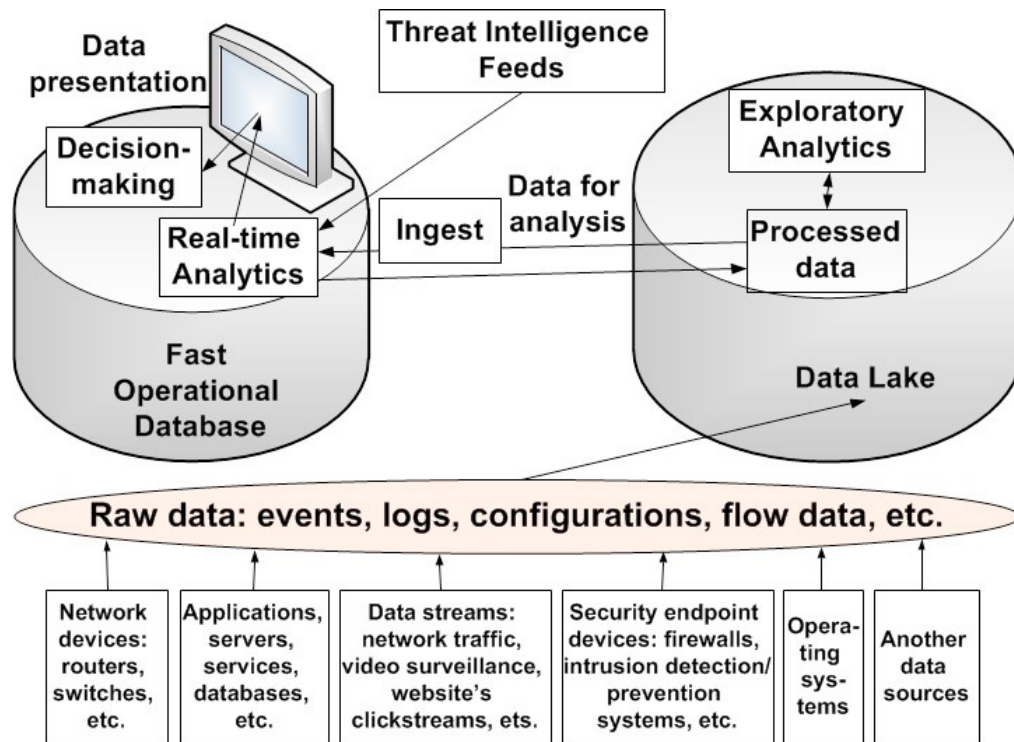


Fig. 1. A simplified NS-related stream data architecture

Threat Intelligence Feeds are a continuous streams of data providing users with constantly updated information about potential network attacks and their sources [8]. They include free and paid Indicator of Compromise (IoC) feeds, bulletins, internal intelligence gathering and strategic partnerships, shared by NS community, including SANS, CERT and others, as well as commercial products, which provide vetted and aggregated data and closed information-sharing communities specific to particular industries or focus areas.

As it is shown in Fig. 1, high-performance real-time analytic systems can enable the integration of data for analysis with earlier processed data from the data lake and raw operational data so that analytics can occur in real-time within the context of any transaction in the network, making these data an intrinsic part of the analytic process. Stream data corresponding to ‘in the moment’ NS-related events from a variety of sources is coming into the data lake. Immediately extracted data preprocessed and ready for analysis is landed in an analytics system, where it is available for analytics.

Currently, volumes of NS-related data is one thing, but the real problem for securing networks' assets is the speed, with which things related to NS happen. That is why even all the NS-related data can be regarded as fast data as it requires immediate corresponding security controls to be activated. The immediate access of the analytics system to the data from the lake means that insights can be discovered in time, to prepare for a further possible course of events and to take appropriate action to reduce risks. Thus, we can use here a fast data concept [4] corresponding to the application of big data analytics to smaller data sets in near-real or real-time to solve a particular problem. Fast data is more emphasized on processing big data streams at speed, and it plays an important role where low latency and high input/output capability for rapid decision updates are required. The goal of fast data analytics is to process immediately structured, semi-structured and unstructured data mined from the data lake so that action can be taken. For that purpose, fast data requires two technologies: a streaming system capable of delivering NS-related events as fast as they come in, and a data store capable of processing each item as fast as it arrives. By using them, fast data processing can be described as “ingest” (get millions of events per second), “decide” (make a data-driven decision on each event) and “analyze in real time” (to enable automated decision-making and provide visibility into operational trends of the events for further prediction).

5. MITRE ATT&CK Matrix as a source for IoCs development

Modern network attacks' types can be described in completely different ways. But it is not effective to develop a separate description for every attack as their number grows every day. That is why a few specifications were designed for unifying these descriptions, such as Incident Object Description and Exchange Format (IODEF) and Real-time Internetwork Defense (RID) by the MILE working group; Intrusion Detection Message Exchange Format (IDMEF) (RFC4765) and IODEF for Structured Cybersecurity Information (IODEF-SCI) (RFC7203) by IETF; Collective Intelligence Framework (CIF) by REN-ISAC; Vocabulary for Event Recording and Incident Sharing (VERIS) by Verizon; Open Indicators of Compromise (OpenIOC) by Mandiant; Open Threat Exchange (OTX) by Alien Vault; Cyber Observable eXpression (CybOX), Trusted Automated eXchange of Indicator Information (TAXII) and Structured Threat Information Expression (STIX) by MITRE. The last two are most often used at present, as they are machine-readable and therefore easily automated.

Next version of STIX – STIX 2.0 is based on the Adversarial Tactics, Techniques & Common Knowledge (ATT&CK) model [9]. Each tactic category from the ATT&CK model contains a list of techniques that an adversary could use to perform that tactic. Here are some examples from the MITRE ATT&CK Matrix, providing a deeper level of granularity in describing what can occur during an attack after the adversary has acquired access that can be used for developing IoCs, further shared by Threat Intelligence Feeds:

1. *Initial Access*: exploitation of public-facing applications, adding hardware, replication through removable media, spearphishing attachment/link, spearphishing via service, compromising supply chain, via trusted relationships, etc.
2. *Execution*: using Command-Line Interface, Control Panel items or dynamic data exchange, through API/GUI/PowerShell/module load, compiled HTML file, scheduled task, scripting, service execution, space after filename, etc.
3. *Persistence*: account manipulation, application shimming, bootkit, browser extensions, change default file association, external remote services, hidden files and directories, hooking, image file execution options injection, launch agent/daemon, login item, logoff, new service, office application startup, path interception, port knocking, port monitor, redundant access, scheduled task, web shell, etc.
4. *Privilege Escalation*: access token manipulation, application shimming, bypass user application control, DLL injection, exploitation for privilege escalation, extra window memory injection, process injection, sudo, etc.
5. *Defense Evasion*: binary padding, clear command history, code signing, DLL search order hijacking, deobfuscate/decode files/information, disabling security tools, indicator blocking/removal, file deletion, file permissions modification, file system logical offsets, Gatekeeper bypass, hidden files/window/users, indirect command execution, install root certificate, masquerading, modify registry, process hollowing, rootkit, timestamp, etc.
6. *Credential Access*: credential dumping, credentials in files, input capture, network sniffing, password filter DLL, two-factor authentication interception, etc.
7. *Discovery*: of system services/information, applications, query registry, network configuration, remote systems, system owner/users, network services/connections, processes, security software, permission groups, files/directories, accounts, peripheral devices, system time, etc.
8. *Lateral Movement*: third-party/application deployment software, remote services, Windows remote management/Admin shares, logon scripts, shared webroot, exploits, remote desktop protocol, replication through removable media, remote file copy, etc.
9. *Collection*: data from local system/removable media/network shared drive, input/audio/video/screen capture, data staged, email collection, automated collection, etc.
10. *Exfiltration*: data compressed/encrypted, exfiltration over other network medium/command and control channel/alternative protocol/physical medium, automated exfiltration, scheduled transfer, data transfer size limits, etc.
11. *Command and Control*: data obfuscation/encoding, fallback channels, custom cryptographic/command & control protocol, multiband communication, commonly/uncommonly used port, standard application/non-application layer protocol, multilayer encryption, connection proxy, communication through removable media, etc.

Another approach for IoCs development can be based on MITRE CAPEC (Common Attack Pattern Enumeration and Classification) [10] where attack patterns are organized hierarchically taking into account 1000 mechanisms that are frequently employed when exploiting a vulnerability. These mechanisms are divided into 9 categories: *Engage in Deceptive Interactions* (156 mechanisms), *Abuse Existing Functionality* (210), *Manipulate Data Structures* (255), *Manipulate System Resources* (262), *Inject Unexpected Items* (152), *Employ Probabilistic Techniques* (223), *Manipulate Timing and State* (172), *Collect and Analyze Information* (118), and *Subvert Access Control* (225).

The data characterizing these phases and their typical consequences as IoCs can be found in all NS-related data collected from different sources in the attacked network. After that it can be extracted and properly correlated so that a particular type of network attack occurred can be determined by a stream data analytics system.

6. Conclusion

Lack of resources and staff's competencies are the most significant barrier in starting to use analytics technology that can perform NS-related data stream processing efficiently and effectively. However, deployment of data streaming tools will not transform organizations' approach to ensuring NS from reactive to proactive overnight. They should start with smaller, more focused efforts to improve organizations' awareness of their "networks' health" in order to better process NS-related data. As they learn their further opportunities to improve their NS strategy with real-time data, they begin to monitor and analyze relationships between events in a particular context. That means that the next step is to tie all data streams collected together. From this information, they will generate new knowledge, further used to adapt for new conditions of their network operations. Hence, our further research is planned in this area.

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